Tracking People In Real Time Video Footage Using Facial Recognition

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***Abstract*— As of now, there is very less knowledge and use of Facial Recognition System for security surveillance in India. This project proposes a system which will use Facial Recognition to track or search a target person from a real time video feed, like a video feed from a surveillance system. Firstly, the system is provided with a Live Video footage of the area that has to be scanned. Then it is provided with an input data set of images of a targeted person for example, a missing person, criminal, etc. Once the input is provided the system will extract a predefined set of facial characteristics from the Input Dataset and create a training module which will help in searching the person from the real-time video footage. If a match is found, the system will identify and mark the person. Also one of the main objectives of this project is to develop the above-mentioned system in conjunction with the existing Surveillance system i.e. to make it compatible with the already installed surveillance cameras in-order keep the costs and hassle of running it at a minimum. The applications of this proposed system can be in Government Organizations like Police, Military, Municipal Corporations, Large Companies, etc for tracking people.**

Keywords— Facial Detection; Facial Recognition; Real Time Video; People Tracking; Intruder Detection.

I. INTRODUCTION

Since 10 years or so confront acknowledgment took an extraordinary development in PC inquire about field and furthermore shaped a standout amongst the best use of picture examination and comprehension. Face acknowledgment innovation measures and matches the striking attributes of a man for the motivations behind recognizable proof or confirmation, utilizing one of a kind facial qualities as it were. Facial acknowledgment is an innovation which has risen as a striking response to address many present day requirements for recognizable proof and the check of personality claims. It unites the guarantee of the more well-known usefulness of visual reconnaissance in the biometric frameworks. Face checking biometric innovation is amazingly flexible and this can be seen by its colossal capability of its current and forthcoming applications. Facial sweep is a viable biometric property/pointer. Different other biometric pointers are utilized for various types of ID applications because of their varieties like exactness, cost, and detecting ability. Face acknowledgment resembles each other PC application depends on particular calculations. Whichever calculation utilized right off the bat recognizes facial highlights by separating points of interest, or highlights, from a picture of the subject's face. For instance, a calculation may investigate the position, measure, state of the eyes, nose, cheekbones and jaw. These highlights are then used to look for different pictures with required coordinating highlights. Alternate counts have a fundamental display of face pictures and a short time later pack the face data, simply saving the data in the photo that is used for going up against affirmation, the required picture, and the face data is then differentiated and the face data. The main framework depended on a format coordinating strategy connected to an arrangement of one of a kind facial highlights, giving a kind of packed face portrayal.

Acknowledgment calculations are of two fundamental methodologies Geometric and Photometric. Geometric which looks at the features where photometric is an authentic approach that distils a photo into features and differentiates the features and layouts to discard the differences. Broadly utilized acknowledgment calculations incorporate head part investigation utilizing Eigen faces, straight discriminant examination, versatile bundle chart coordinating utilizing the Fisher confront calculation, the shrouded Markov display, the multi-linear subspace picking up utilizing tensor portrayal, and the neuronal system spurred dynamic connection coordinating framework. The idea of the issue, PC related scientists are intrigued, as well as neuroscientists and clinicians. It is the general supposition that advances in PC vision research will give helpful experiences to neuroscientists and therapists into how human mind functions. There are an extensive number of business, security, and scientific applications requiring the utilization of face acknowledgment innovations. These applications incorporate computerized swarm observation, get to control, mug shot recognizable proof, confront remaking, plan of human PC interface HCI and substance based picture database administration. Various business confront acknowledgment frameworks have been sent, for example, Cognitec, Eyematic, Visage and Identix.

II. LITERATURE SURVEY

Each product which bargains in the human face acknowledgment and handling of the highlights of the human countenances amid the show of differing feelings, requires an earlier inside and out examination into the most able, strong

and proficient method that can be received for the procedure, earlier work if any that has been done in the field and the degree of expound explore led in the region. Impressive work has been done in the range of face discovery and acknowledgment with late concentrate being on the extraction of observation levels [1]. Jitao Sang, Changsheng Xu, propose a worldwide face-name chart coordinating based structure for hearty film character ID. The proposed plot have a place with the worldwide coordinating based classification, where outer content assets are used. To enhance the vigor, the ordinal chart is utilized for face and name diagram portrayal and a novel chart coordinating calculation called Error Correcting Graph Matching (ECGM) is presented [2]. Pantic and Rothkrantz contrived a multi-indicator way to deal with limitation of facial element is utilized to spatially test the profile form and the shapes of the facial segments, for example, the eyes and the mouth. An acknowledgment rate of 86% is accomplished [3]. Rowley et al. propose a neural system based calculation to recognize upright, frontal viewpoints of appearances in dim scale pictures. The calculation works by executing at least one neural systems straightforwardly to sections of the info picture and arbitrating their outcomes. Each system is prepared to distinguish the nearness or nonappearance of a face [4]. Ming Hsuan Yang et al. talk about ways to deal with distinguish faces in pictures and the motivation behind their paper is to order and assess these methodologies, recognize their impediments and propose healing measures: Knowledge based, Feature Invariant, Template Matching and Appearance based methodologies lodging calculations, for example, PCA, Neural Networks and so forth. [5].Shahrin Azuan Nazeer et al. talk about that current techniques have been upright with building a hearty framework that spotlights on both portrayal and acknowledgment utilizing fake Neural Networks. It at that point assesses the execution of the framework by applying two photometric standardization methods: histogram adjustment and homomorphic separating and contrasting and Euclidean Distance and Normalized Correlation classifiers [6]. Steve Lawrence et al. consolidates nearby picture inspecting, a self-arranging map neural system and a convolutional neural system keeping in mind the end goal to recognize two
general classifications of face acknowledgment: We wish to find a man from a vast database of appearances (e.g., in a police database), OR, we wish to recognize singular people continuously (e.g., in a security checking framework) [7]. L. Mama and K. Khorasani present a method for outward appearance acknowledgment, which fuses the two-dimensional discrete cosine change over the whole picture as an element indicator and a productive one-concealed layer feed forward neural system to order outward appearance. The neural system based acknowledgment techniques are observed to be especially encouraging [8], [9], since the neural systems can without much of a stretch actualize the mapping from the element space of face pictures to the outward appearance space [10]. Petar S. Aleksic et al. display a programmed multistream HMM outward appearance acknowledgment framework and find out that its execution is outstandingly enhanced by demonstrating the trustworthiness of various floods of outward appearance data using multi-stream shrouded Markov Models (HMMs). The proposed framework uses facial liveliness parameters (FAPs), upheld by the MPEG-4 standard, as highlights for outward appearance arrangement.

III.PROPOSED SYSTEM

In perspective of the foundation of related work on Face Recognition and Expressions Analysis utilizing different calculations depicted over, the creators have possessed the capacity to propose a broad utilization of Neural Network for identifying and perceiving faces and additionally investigating outward appearances in changed fields. It is a compelling model for highlight identification and investigation in faces giving us high exactness and promising outcomes when utilized on an expansive scale.

The Proposed System will utilize Facial Recognition to track or hunt an objective individual from an ongoing video sustain, similar to a video encourage from an observation framework. Right off the bat, the framework is given a Live Video film of the territory that must be filtered. At that point it is given an info informational index of images of a focused on individual for instance, a missing individual, criminal, and so forth. Once the information is given the framework will separate a predefined set of facial attributes from the Input Dataset and make a preparation module which will help in looking through the individual from the ongoing video film. On the off chance that a match is discovered, the framework will distinguish and check the individual.

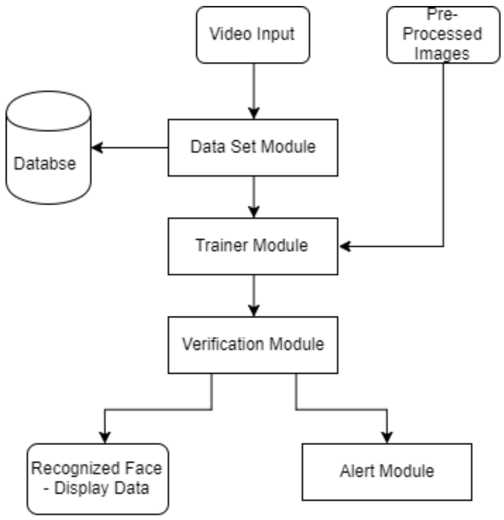

Figure 1. Modules Present In The Proposed System

In order to describe the work being done by us, it has been broadly divided into four modules:

Dataset Module: It is an automated mechanism that scan and captures images of a living person’s characteristics. A pre-processing module locates Eye position, surrounding lighting condition and color variance. Firstly the presence of a face must be detected. Different face recognition approaches can be used to extract facial characteristics. The output of the Analysis is packaged into a set which is called as the Analyzed Data.

Database Module: It is an object which looks after gathering, processing, storage and compression of captured data with local data.

Verification Module: It interfaces with the application system. It matches the characteristics of the captured data with the database. A pre-processed pattern is compared with each face print in the database. After the comparisons are made the system assigns a value from 1-10 & if score is above predefined threshold a match is declared and the details from the database are displayed.

- Working Of The Proposed System

Step 1: A live video footage is provided to the system , and the initial step would be to preprocess the video, like conversion, transition and framing.

Step 2. If a face is present, the eye coordinates of the face shall first to be extracted and normalized.

Step 3: A Fudicial Marker is places on the frames which are being used in the process to facilitate further markings.

Step 4: A Face Bunch Graph is generated and overlayed on the Fudicial points marked.

Step 5: A Gabor Filter is used to do the texute analysis and the data is combined with face bunch graph to extract the facial characteristics.

Step 6: The facial characteristics pertaining to the 80 nodal points are extracted.

Step 7: These extracted characteristics of the person are then matched to that of the second image or database.

Step 8: After the comparisons are made the system assigns a value from 1-10 & if score is above predefined threshold a match is declared and the data for the matching person is fetched from the database.

Step 9: A square around the person face is marked and the data from the database of the matching person is displayed.If the person is unknown a marker called “Unknown Person” is displayed.

Step 10: An alert is sent to the admin to notify an unknown person is being tracked in the frame.

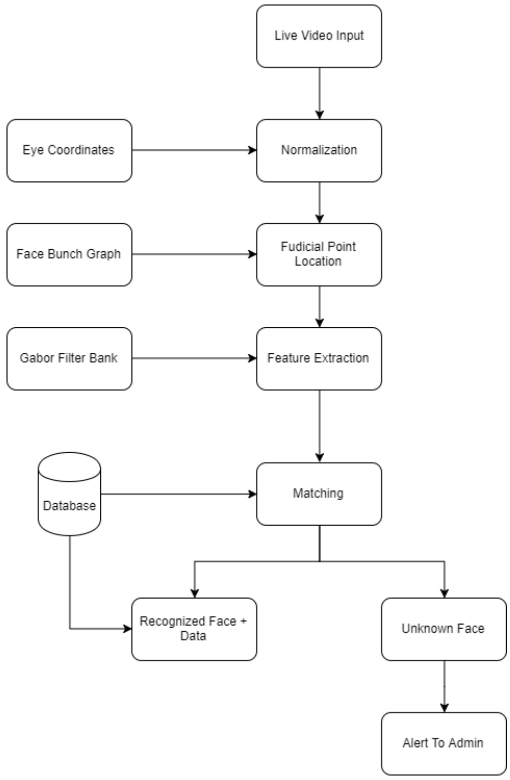

Figure 2. Flowchart Of Working Of System

- Algorithm

Elastic Bunch Graph Algorithm is being used in this project, it is a popular algorithm used in the computer vision and object detection domain. It creates a graph based on what is see and uses different nodal points.

- For Face Recognition In Ideal Pose:

Stage 1: Building a face chart. The initial step to bootstrap the framework is to characterize the diagram structure for the given stance. Subsequently, we take the main picture and physically characterize hub areas on the face that are anything but difficult to confine, for example, the edges of the eyes or mouth, the focal point of the eyes, the tip of the commotion, a few focuses on the blueprint and so on. We additionally characterize edges between the hubs. This constitutes our first face chart.

Stage 2: Building a face pack chart. The single face chart characterized above can be seen as a cluster diagram with only one example in it. It can be coordinated onto the second face picture, yet in the event that the initial two face pictures are not fundamentally the same as the match is of low quality. For example, the tip-of-the-nose hub may be put at the cheek, so we have to move the hub onto the tip of the nose by hand. After some such manual adjustment, the chart is satisfactory and constitutes the second case in the bundle diagram. The cluster diagram with two occurrences is then coordinated onto the third picture, and after some manual rectification, we have a third case for the bundle chart. By rehashing this procedure, the cluster diagram develops, and as it develops the match onto new pictures gets increasingly dependable. On the off chance that we are happy with the nature of the matches and just minimal manual adjustment is required, we are finished with building the pack diagram. Let say this occurs in the wake of having prepared the initial 100 pictures.

Stage 3: Building the model exhibition of diagrams. Since we now have a pack chart that gives adequate quality in finding the hub areas in another face, we can process the staying 900 pictures completely naturally. To keep away from the qualification between physically redressed and completely naturally produced demonstrate charts, one can make the pack diagram on an additional arrangement of 100 pictures unmistakable from the 1000 model pictures. At that point, each of the 1000 model charts can be made naturally. We are presently in the position to perform confront acknowledgment on another test picture.

Stage 4: Building the test diagram. Expect we are given another picture and should locate the delineated individual in the display. To begin with we have to make a chart for the test picture. This procedure works precisely with respect to the model pictures, only that we utilize the test picture.

Stage 5: Comparison with every single model diagram. The picture diagram is contrasted and every single model chart, bringing about 1000 comparability esteems. These frame the premise of the acknowledgment choice. Notice this does not require EBGM any longer, just the charts are contrasted concurring with the similitude work (3).

Stage 6: Recognition. For acknowledgment clearly, the model chart with the most astounding closeness with the picture diagram is the possibility to be perceived. In any case, if the best likeness esteem is moderately low, the framework may choose that the individual on the test picture is not in the model exhibition by any means; and if there are more than one high similitude esteems, the framework may choose that the individual is in all probability in the model display however that there are a few conceivable hopefuls. Just if the most elevated comparability is high and the following one is low can the framework perceive the face on the test picture with high unwavering quality.

- For Face Recognition In Different Poses:

EBGM is vigorous to little posture changes. Bigger contrasts in posture adjust the neighborhood includes excessively to correspond focuses to be discovered, a few highlights are impeded in a few stances. This trouble can be overwhelmed by making bundle charts for various postures and coordinating every one of them to the test picture. As little posture contrasts are endured by the coordinating procedure, just coarse examining of stance is important. Just if the postures coordinated are generally equivalent, high comparability esteems and great correspondences will come about.

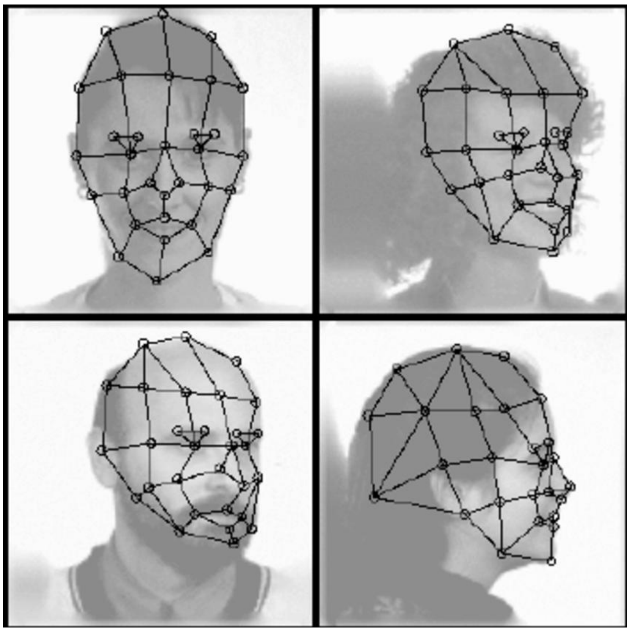

Figure 3. Fudicial Point Extraction and Face Bunch Graph Generation

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The execution of the proposed system is implemented in Python 2.7 using OpenCV libraries for the face detection & recognition developed by Intel. The default process uses Haar Cascades designed by Intel, in the proposed system we use the modified version of Haar Cascades for classifying the faces. The default version can only be used for frontal face classification and identification. In the proposed system, the Haar Cascades have been specifically retrained in order to recognize faces at various angles, now the detection and recognition process uses these modified Cascades thus returning a better accuracy in the detection process of the faces.

The following table represents the Accuracy of Detection and Recognition at various levels:

False Acceptance Rate (FAR) (float[f])	False Acceptance Rate (FAR) %	Matching Accuracy (approx. %)	Recognition Rate (%)
0.01	1	60	99
0.005	0.5	70	90
0.001	0.1	85	85
0.0005	0.05	95	70
0.00001	0.001	98	40
0.00005	0.005	99	10

Table1. Table representing FAR in conjunction with Accuracy Of Detection & Recognition Rate

Below is the comparison graph of various results obtained from the default Haar Cascades, and that of the proposed system which uses the modified Haar Cascades.

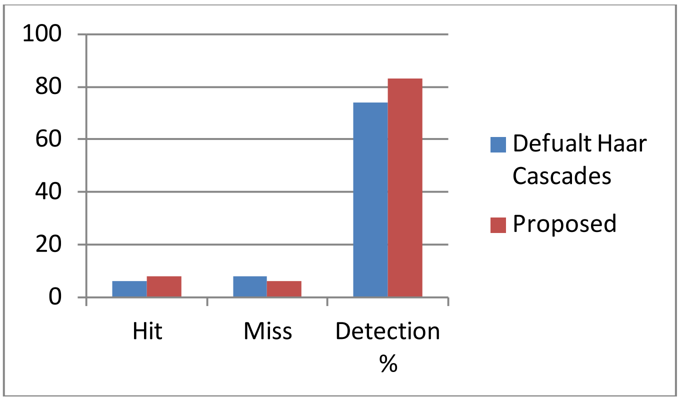

Figure 4. Comparison Of Default Haar Cascades and Proposed system.

Figure 5. Output of the current execution of proposed system.

V.CONCLUSION

It has been certainly made out from the result and analysis that the proposed work has provided a better implementation of the current algorithms with integration of the modified work. The proposed work of tracking people in live video is just small part of the much larger picture. Compared to legacy systems, assay of facial expressions can help expedite a lot of manual labor that goes into the respective area of work, unnecessarily. Similar approach can be used to further develop system which detect crimes, identify mood of people, security systems, etc. Live Subject tracking, Facial Recognition and Analysis is a much deeper subject which provides immense research opportunities and also development for the greater good of the society.

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